

H2 Standard integrals

Name: _____

Class: _____

Date: _____

Time: **354 minutes**

Marks: **294 marks**

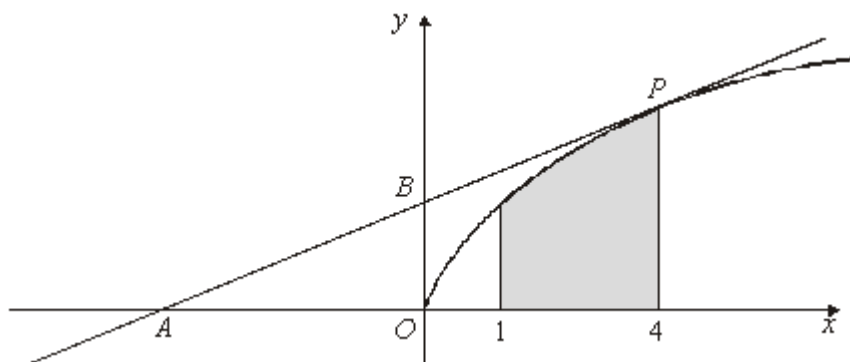
Comments:

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Q1.

The diagram shows a curve C with equation $y = \sqrt{x}$. The point O is the origin $(0, 0)$.



The region bounded by the curve C , the x -axis and the vertical lines $x = 1$ and $x = 4$ is shown shaded in the diagram.

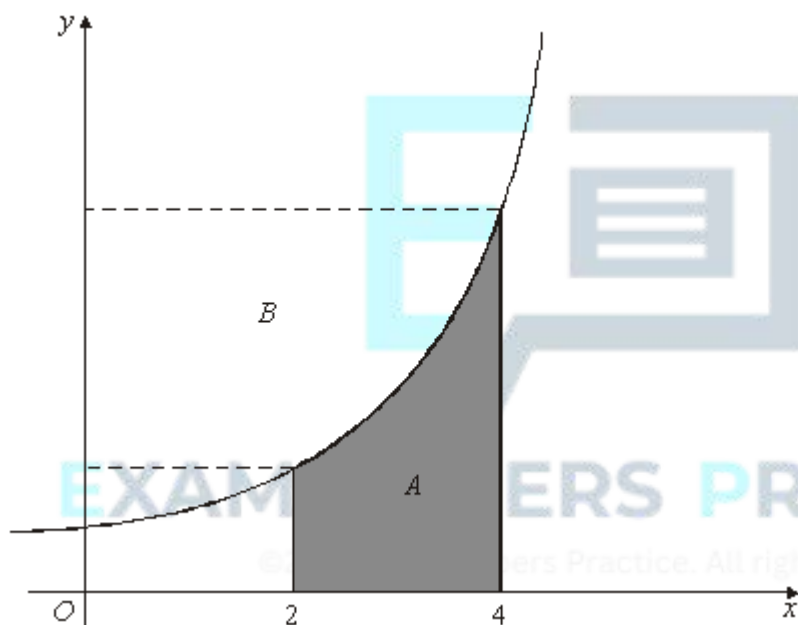
- (a) (i) Write $\sqrt{x}$ in the form x^p , where p is a constant. (1)
 - (ii) Find $\int \sqrt{x} \, dx$. (2)
 - (iii) Hence find the area of the shaded region. (3)
 - (b) The point on C for which $x = 4$ is P . The tangent to C at the point P intersects the x -axis and the y -axis at the points A and B respectively.
 - (i) Find an equation for the tangent to the curve C at the point P . (4)
 - (ii) Find the area of the triangle AOB . (3)
 - (c) Describe the single geometrical transformation by which the curve with equation $y = \sqrt{x-1}$ can be obtained from the curve C . (2)
 - (d) Use the trapezium rule with four ordinates (three strips) to find an approximation for $\int_1^4 \sqrt{x-1} \, dx$, giving your answer to three significant figures. (4)
- (Total 19 marks)**

Q2.

- (a) Find $\int e^{4x} dx$. (1)
- (b) Use integration by parts to find $\int e^{4x}(2x+1) dx$. (3)
- (c) By using the substitution $u = 1 + \ln x$, or otherwise, find $\int \frac{1 + \ln x}{x} dx$. (4)
- (Total 8 marks)**

Q3.

The diagram shows part of the graph of $y = e^{2x} + 3$.



- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} + 3$. (4)
- (b) Find the exact value of the area of the shaded region A . (4)
- (c) The region B is indicated on the diagram.

Find the area of the region B , giving your answer in the form $pe^8 + qe^4$, where p and q are numbers to be determined.

(4)
(Total 12 marks)

Q4.

- (a) Express $\frac{3x-5}{(x+3)(2x-1)}$ in the form $\frac{A}{x+3} + \frac{B}{2x-1}$. (3)

- (b) Hence find $\int \frac{3x-5}{(x+3)(2x-1)} dx$. (3)
- (Total 6 marks)**

Q5.

- (a) Express $\sin 2x$ in terms of $\sin x$ and $\cos x$. (1)

- (b) Using the identity $\cos(A+B) = \cos A \cos B - \sin A \sin B$:

- (i) express $\cos 2x$ in terms of $\sin x$ and $\cos x$; (2)

- (ii) show, by writing $3x$ as $(2x+x)$, that

$$\cos 3x = 4 \cos^3 x - 3 \cos x$$

(4)

- (c) Show that $\int_0^{\frac{\pi}{2}} \cos^3 x \, dx = \frac{2}{3}$. (5)
- (Total 12 marks)**

Q6.

A particle, of mass 10 kg, moves on a smooth horizontal plane. At time t seconds, the acceleration of the particle is given by

$$\{(40t + 3t^2)\mathbf{i} + 20e^{-4t}\mathbf{j}\} \text{ m s}^{-2}$$

where the vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

- (a) At time $t = 1$, the velocity of the particle is $(6\mathbf{i} - 5e^{-4}\mathbf{j}) \text{ m s}^{-1}$.

Find the velocity of the particle at time t . (5)

- (b) Calculate the initial speed of the particle. (3)
- (Total 8 marks)**

Q7.

A car accelerates from rest along a straight horizontal road.

The car's engine produces a constant horizontal force of magnitude 4000 N.

At time t seconds, the speed of the car is $v \text{ m s}^{-1}$, and a resistance force of magnitude $40v$ newtons acts upon the car.

The mass of the car is 1600 kg.

(a) Show that $\frac{dv}{dt} = \frac{100 - v}{40}$.

(2)

(b) Find the velocity of the car at time t .

(6)

(Total 8 marks)

Q8.

A particle, of mass 50 kg, moves on a smooth horizontal plane. A single horizontal force

$$[(300t - 60t^2)\mathbf{i} + 100e^{-2t}\mathbf{j}] \text{ newtons}$$

acts on the particle at time t seconds.

The vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular unit vectors.

(a) Find the acceleration of the particle at time t .

(2)

(b) When $t = 0$, the velocity of the particle is $(7\mathbf{i} - 4\mathbf{j}) \text{ m s}^{-1}$.

Find the velocity of the particle at time t .

(4)

(c) Calculate the speed of the particle when $t = 1$.

(4)

(Total 10 marks)

Q9.

Alice places a toy, of mass 0.4 kg, on a slope. The toy is set in motion with an initial velocity of 1 m s^{-1} down the slope. The resultant force acting on the toy is $(2 - 4v)$ newtons, where $v \text{ m s}^{-1}$ is the toy's velocity at time t seconds after it is set in motion.

(a) Show that $\frac{dv}{dt} = -10(v - 0.5)$.

(2)

- (b) By using $\int \frac{1}{v-0.5} dv = - \int 10 dt$, find v in terms of t . (5)
- (c) Find the time taken for the toy's velocity to reduce to 0.55 m s^{-1} . (3)
- (Total 10 marks)

Q10.

A particle moves in a straight line. At time t seconds, it has velocity $v \text{ m s}^{-1}$, where

$$v = 6t^2 - 2e^{-4t} + 8$$

and $t \geq 0$.

- (a) (i) Find an expression for the acceleration of the particle at time t . (2)
- (ii) Find the acceleration of the particle when $t = 0.5$. (2)
- (b) The particle has mass 4 kg .
Find the magnitude of the force acting on the particle when $t = 0.5$. (1)
- (c) When $t = 0$, the particle is at the origin.
Find an expression for the displacement of the particle from the origin at time t . (4)
- (Total 9 marks)

Q11.

Vicky has mass 65 kg and is skydiving. She steps out of a helicopter and falls vertically. She then waits a short period of time before opening her parachute. The parachute opens at time $t = 0$ when her speed is 19.6 m s^{-1} , and she then experiences an air resistance force of magnitude $260v$ newtons, where $v \text{ m s}^{-1}$ is her speed at time t seconds.

- (a) When $t > 0$:
- (i) show that the resultant downward force acting on Vicky is $65(9.8 - 4v)$ newtons (1)
- (ii) show that $\frac{dv}{dt} = -4(v - 2.45)$. (2)

- (b) By showing that $\int \frac{1}{v - 2.45} dv = - \int 4 dt$, find v in terms of t .

(5)

(Total 8 marks)

Q12.

A particle moves in a horizontal plane under the action of a single force, $\mathbf{F}$ newtons. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively. At time t seconds, the velocity of the particle, $\mathbf{v} \text{ m s}^{-1}$, is given by

$$\mathbf{v} = 4e^{-2t} \mathbf{i} + (6t - 3t^2) \mathbf{j}$$

- (a) Find an expression for the acceleration of the particle at time t .

(3)

- (b) The mass of the particle is 5 kg.

- (i) Find an expression for the force $\mathbf{F}$ acting on the particle at time t .

(2)

- (ii) Find the magnitude of $\mathbf{F}$ when $t = 0$.

(2)

- (c) Find the value of t when $\mathbf{F}$ acts due west.

(2)

- (d) When $t = 0$, the particle is at the point with position vector $(6\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 14 marks)

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Q13.

A car, of mass $m \text{ kg}$, is moving along a straight horizontal road. At time t seconds, the car has speed $v \text{ m s}^{-1}$. As the car moves, it experiences a resistance force of magnitude $2mv^{\frac{5}{4}}$ newtons. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -2v^{\frac{5}{4}}$$

(1)

- (b) The initial speed of the car is 16 m s^{-1} .

Show that

$$v = \left(\frac{2}{t+1} \right)^4$$

(5)

(Total 6 marks)

Q14.

A golf ball, of mass m kg, is moving in a straight line across smooth horizontal ground. At time t seconds, the golf ball has speed v m s⁻¹. As the golf ball moves, it experiences

a resistance force of magnitude $0.2mv^{\frac{1}{2}}$ newtons until it comes to rest. No other horizontal force acts on the golf ball.

Model the golf ball as a particle.

(a) Show that

$$\frac{dv}{dt} = -0.2v^{\frac{1}{2}}$$

(1)

(b) When $t = 0$, the speed of the golf ball is 16 m s⁻¹.

Show that $v = (4 - 0.1t)^2$.

(5)

(c) Find the value of t when $v = 1$.

(3)

(d) Find the distance travelled by the golf ball as its speed decreases from 16 m s⁻¹ to 1 m s⁻¹.

(4)

(Total 13 marks)

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Q15.

A particle is moving along a straight line. At time t , the velocity of the particle is v .

$$-\frac{\lambda}{v^{\frac{1}{4}}}$$

The acceleration of the particle throughout the motion is $-\frac{\lambda}{v^{\frac{1}{4}}}$, where λ is a positive constant. The velocity of the particle is u when $t = 0$.

Find v in terms of u , λ and t .

(Total 7 marks)

Q16.

A particle moves along a straight line. At time t , it has velocity v , where

$$v = 4t^3 - 8 \sin 2t + 5$$

When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(Total 4 marks)

Q17.

A stone, of mass 0.05 kg, is moving along the smooth horizontal floor of a tank, which is filled with oil. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude $0.08 v^2$.

(a) Show that

$$\frac{dv}{dt} = -1.6v^2$$

(2)

(b) The initial speed of the stone is 3 m s^{-1} .

Show that

$$v = \frac{15}{5 + 24t}$$

(5)

(Total 7 marks)

Q18.

A particle moves in a straight line and at time t it has velocity v , where

$$v = 3t^2 - 2 \sin 3t + 6$$

(a) (i) Find an expression for the acceleration of the particle at time t .

(2)

(ii) When $t = \frac{\pi}{3}$, show that the acceleration of the particle is $2\pi + 6$.

(2)

(b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 8 marks)

Q19.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has

speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s^{-1} .

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s^{-1} .

(3)

- (d) Find, in terms of m , the work done by the force in slowing the car from 20 m s^{-1} to 10 m s^{-1} .

(3)

(Total 11 marks)

Q20.

A car, of mass m , is moving along a straight smooth horizontal road. At time t , the car has speed v . As the car moves, it experiences a resistance force of magnitude $0.05mv$. No other horizontal force acts on the car.

- (a) Show that

$$\frac{dv}{dt} = -0.05v$$

(1)

- (b) When $t = 0$, the speed of the car is 20 m s^{-1} .

Show that $v = 20e^{-0.05t}$.

(4)

- (c) Find the time taken for the speed of the car to reduce to 10 m s^{-1} .

(3)

(Total 8 marks)

Q21.

A particle has mass 800 kg . A single force of $(2400 \mathbf{i} - 4800t \mathbf{j})$ newtons acts on the particle at time t seconds. No other forces act on the particle.

- (a) Find the acceleration of the particle at time t .

(2)

- (b) At time $t = 0$, the velocity of the particle is $(6\mathbf{i} + 30\mathbf{j}) \text{ m s}^{-1}$. The velocity of the particle at time t is $\mathbf{v} \text{ m s}^{-1}$.

Show that

$$\mathbf{v} = (6 + 3t)\mathbf{i} + (30 - 3t^2)\mathbf{j} \quad (4)$$

- (c) Initially, the particle is at the point with position vector $(2\mathbf{i} + 5\mathbf{j}) \text{ m}$.

Find the position vector, $\mathbf{r}$ metres, of the particle at time t .

(5)

(Total 11 marks)

Q22.

A stone of mass m is moving along the smooth horizontal floor of a tank which is filled with a viscous liquid. At time t , the stone has speed v . As the stone moves, it experiences a resistance force of magnitude λmv , where λ is a constant.

- (a) Show that

$$\frac{dv}{dt} = -\lambda v$$

(2)

- (b) The initial speed of the stone is U .

Show that

$$v = Ue^{-\lambda t}$$

(4)

(Total 6 marks)

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Q23.

A particle moves in a straight line and at time t has velocity v , where

$$v = 2t - 12e^{-t}, \quad t \geq 0$$

- (a) (i) Find an expression for the acceleration of the particle at time t .

(2)

- (ii) State the range of values of the acceleration of the particle.

(3)

- (b) When $t = 0$, the particle is at the origin.

Find an expression for the displacement of the particle from the origin at time t .

(4)

(Total 9 marks)

Q24.

A student is modelling the motion of a small boat as it moves on a lake. When the speed of the boat is 12 m s^{-1} , the engine is switched off. At time t seconds later, it has a velocity of $v \text{ m s}^{-1}$ and experiences a resistance force of magnitude $20v$ newtons. The mass of the boat is 80 kg .

To set up a simple model for the motion of the boat, the student assumes that the water in the lake is still and that the boat travels in a straight line.

- (a) Explain how these two assumptions allow the student to create a simple model.

(2)

- (b) State one other assumption that the student should make.

(1)

- (c) (i) Express $\frac{dv}{dt}$ in terms of v .

(2)

- (ii) Find an expression for v in terms of t .

(5)

(Total 10 marks)

Q25.

A particle of mass 20 kg moves along a straight horizontal line. At time t seconds the velocity of the particle is $v \text{ m s}^{-1}$. A resistance force of magnitude $10\sqrt{v}$ newtons acts on the particle while it is moving. At time $t = 0$ the velocity of the particle is 25 m s^{-1} .

- (a) Show that, at time t

$$v = \left(\frac{20 - t}{4} \right)^2$$

(7)

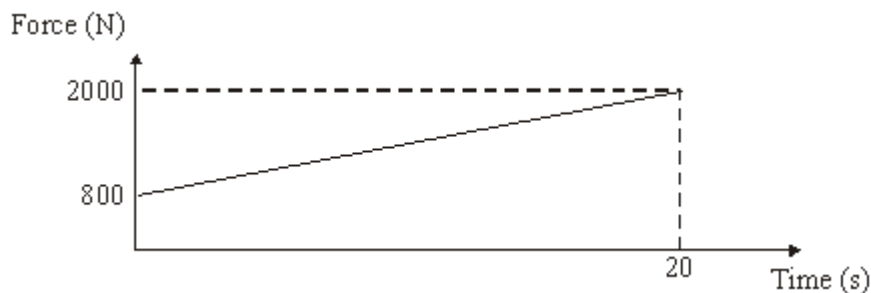
- (b) State the value of t when the particle comes to rest.

(1)

(Total 8 marks)

Q26.

The graph shows a model for the resultant horizontal force on a car, which varies as it accelerates from rest for 20 seconds. The mass of the car is 1200 kg .



- (a) The acceleration of the car at time t seconds is $a \text{ m s}^{-2}$. Show that

$$a = \frac{2}{3} + \frac{t}{20}, \text{ for } 0 \leq t \leq 20 \quad (5)$$

- (b) Find an expression for the velocity of the car at time t . (3)

- (c) Find the distance travelled by the car in the 20 seconds. (4)

- (d) An alternative model assumes that the resultant force increases uniformly from 900 to 2100 newtons during the 20 seconds. Which term in your expression for the velocity would change as a result of this modification? Explain why. (2)

(Total 14 marks)

Q27.

A particle travels along a straight line. At time t , the velocity of the particle is v . When $t = 0$,

$v = u$. The acceleration of the particle throughout the motion is $\frac{k}{v}$, where k is a constant.

Show that

$$v^2 = u^2 + 2kt$$

(Total 6 marks)

Q28.

John drops a rock of mass m kg from the top of a vertical cliff. At time t seconds after the rock is dropped, it has speed $v \text{ m s}^{-1}$ and the air resistance to its motion is of magnitude $0.1mv$ newtons.

- (a) Show that the differential equation connecting v and t for this motion is $\frac{dv}{dt} = g - 0.1v$. (2)

(b) Solve this differential equation to find v in terms of g and t . (5)

(c) Hence show that the speed of the rock cannot exceed 98 m s^{-1} . (1)

(Total 8 marks)

Q29.

A car accelerates from rest along a straight road for 5 seconds. At time t seconds, its acceleration, $a \text{ m s}^{-2}$, is given by

$$a = t - \frac{t^2}{5}$$

(a) By integrating, find an expression for the velocity of the car at time t . (3)

(b) Find the velocity of the car at the end of the 5 second period. (2)

(c) Find the distance travelled by the car during the 5 second period. (4)

(Total 9 marks)

Q30.

A particle moves on a straight line. At time t seconds its acceleration, $a \text{ m s}^{-2}$ is given by

$$a = 20 \sin 4t$$

(a) Initially the particle is at rest. Find an expression for the velocity of the particle at time t . (4)

(b) Initially the displacement of the particle from the origin is 0.8 metres. Find an expression for the displacement of the particle at time t . (4)

(Total 8 marks)

Q31.

A particle has mass 2000 kg. A single force, $\mathbf{F} = 1000\mathbf{i} - 5000\mathbf{j}$ newtons, acts on the particle, at time t seconds. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are perpendicular. No other forces act on the particle.

(a) Find an expression for the acceleration of the particle. (2)

(b) At time $t = 0$, the velocity of the particle is $6\mathbf{j} \text{ m s}^{-1}$. Show that at time t the velocity, $\mathbf{v}$

m s^{-1} , of the particle is given by

$$\mathbf{v} = \frac{t^2}{4}\mathbf{i} + \left(6 - \frac{5t}{2}\right)\mathbf{j}$$

(4)

- (c) The particle is initially at the origin. Find an expression for the position vector, $\mathbf{r}$ metres, of the particle at time t seconds.

(4)

(Total 10 marks)

Q32.

A particle moves along a straight line. At time t seconds the acceleration, $a \text{ m s}^{-2}$, of the particle is given by

$$a = 2 - 2e^{-t}$$

At time $t = 0$ the particle is at the origin moving with a velocity of 4 m s^{-1} .

- (a) Show that the velocity, $v \text{ m s}^{-1}$, at time t seconds is given by

$$v = 2t = 2e^{-t} + 2$$

(3)

- (b) Find an expression for the distance of the particle from the origin at time t .

(4)

(Total 7 marks)